STATISTICAL ANALYSIS PLAN

Protocol ID: NSCLC-IO-001
SAP Version: 1.0
Indication: Advanced NSCLC
Study Phase: Phase II Randomized Study

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1. Introduction

This Statistical Analysis Plan (SAP) describes the statistical methodology for evaluating the efficacy and safety of Immunotherapy A compared with Standard of Care (SOC) in patients with advanced non-small cell lung cancer (NSCLC).

The SAP is developed in accordance with the study protocol and follows the estimand framework described in ICH E9(R1), ensuring alignment between clinical objectives, data handling, and statistical analysis.

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2. Study Design

This is a Phase II, randomized, open-label study in patients with advanced NSCLC.

Patients will be randomized 1:1 to:

- Immunotherapy A
- Standard of Care

Randomization will be stratified by PD-L1 expression:

- <1%
- 1–49%
- ≥50%

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3. Objectives and Endpoints

3.1 Primary Objective

To compare Objective Response Rate (ORR) between treatment arms.

Primary Endpoint

ORR defined as the proportion of patients achieving:

- Complete Response (CR) or
- Partial Response (PR)

as assessed per RECIST 1.1.

3.2 Secondary Objectives

- Evaluate Progression-Free Survival (PFS)
- Assess treatment effect across PD-L1 subgroups

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4. Estimands

4.1 Primary Estimand (ORR)

- **Population:** All randomized patients (Intent-to-Treat)
- **Variable:** Binary response (Responder vs Non-responder)

Intercurrent Events (ICE) Handling

ICE	Strategy
Treatment discontinuation	Treatment Policy
New anti-cancer therapy	Treatment Policy
Death before response	Composite (non-responder)

Summary Measure

- Odds Ratio (OR)
- Risk Difference

4.2 Sensitivity Estimand (Hypothetical)

This estimand evaluates treatment effect under the assumption that patients did not initiate subsequent anti-cancer therapy.

- Missing outcomes will be handled using **Multiple Imputation (MI)**
- Imputation model will include:
    - Treatment
    - PD-L1 group
    - Baseline covariates

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5. Analysis Populations

- **Full Analysis Set (FAS / ITT):** All randomized patients
- **Per Protocol Set:** Patients without major protocol deviations
- **Safety Set:** All treated patients

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6. Statistical Methods

6.1 Primary Analysis (ORR)

The primary endpoint will be analyzed using logistic regression:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \cdot Treatment + \beta_2 \cdot PDL1$$

Where:

- p= probability of response

The model will include:

- Treatment arm
- PD-L1 stratification

Outputs

- Odds Ratio
- 95% Confidence Interval
- Two-sided p-value ($\alpha = 0.05$)

6.2 Secondary Analysis (PFS)

PFS will be analyzed using:

Kaplan–Meier Estimation

- Median PFS
- Survival curves

Cox Proportional Hazards Model

$$h(t) = h_0(t)\exp(\beta_1 \cdot Treatment)$$

Additional Analysis

If proportional hazards assumption is violated:

- **Restricted Mean Survival Time (RMST)** will be used

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7. Subgroup Analysis

Treatment effect will be evaluated across PD-L1 subgroups:

- <1%
- 1–49%
- ≥50%

Results will be presented using:

- Forest plots
- Interaction tests

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8. Bayesian Subgroup Analysis

A Bayesian hierarchical model will be used to improve precision in subgroup estimates.

$$\theta_j \sim \mathcal{N}(\mu, \tau^2)$$

Where:

- θ_j = subgroup-specific log-odds
- τ^2 = between-group variance

Priors

- $\mu \sim \mathcal{N}(0,5)$
- $\tau \sim$ Half-Cauchy$(0,2.5)$

Inference

- MCMC sampling
- Convergence criteria:
  - R-hat < 1.01
  - Effective Sample Size > 1000

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9. Missing Data Handling

Primary Analysis

- Missing response → Non-responder

Sensitivity Analyses

- Multiple Imputation (MAR assumption)
- Tipping-point analysis

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10. Multiplicity

Multiplicity will be controlled using a hierarchical testing strategy:

1. ORR tested first
2. If significant → PFS tested

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11. Interim Analysis

A futility analysis will be conducted using an O'Brien–Fleming boundary.

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12. Tables, Listings, and Figures (TLFs)

The analysis outputs will include tables, listings, and figures supporting the primary and secondary endpoints.

A separate TLF document will be generated to present detailed results.

Planned outputs include:

- Table 14.1.1: ORR Summary
- Figure 14.2.1: Kaplan–Meier Curve (PFS)
- Figure 14.2.2: Forest Plot (Subgroups)
- Figure 14.2.3: Bayesian Posterior Distribution

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13. Statistical Considerations and Rationale

13.1 Choice of Primary Endpoint

Objective Response Rate (ORR) is selected as the primary endpoint for this Phase II study to enable early assessment of anti-tumor activity. In oncology drug development, ORR is commonly used in early-phase trials to provide a rapid signal of efficacy, particularly for immunotherapy agents.

While Progression-Free Survival (PFS) provides a more comprehensive assessment of treatment benefit, it typically requires longer follow-up and may be influenced by post-progression therapies. Therefore, PFS is included as a key secondary endpoint to support the durability of response observed in ORR.

13.2 Estimand Strategy and Clinical Relevance

The primary estimand is based on a **treatment policy strategy**, which reflects the treatment effect regardless of treatment discontinuation or initiation of subsequent anti-cancer therapy.

This approach is selected to:

- Reflect real-world clinical practice
- Capture the overall treatment effect in the presence of intercurrent events

A **hypothetical estimand** is included as a sensitivity analysis to evaluate treatment effect under the assumption that patients did not initiate subsequent therapy. This provides insight into the intrinsic efficacy of the investigational treatment, independent of post-baseline interventions.

The combination of treatment policy and hypothetical estimands ensures a comprehensive understanding of both effectiveness and efficacy.

13.3 Subgroup Strategy and Precision Considerations

PD-L1 expression is a clinically relevant biomarker in NSCLC and is used for stratification and subgroup analysis. However, subgroup analyses are often limited by small sample sizes, leading to imprecise estimates and increased variability.

To address this, a **Bayesian hierarchical modeling approach** is employed to borrow information across PD-L1 subgroups. This approach:

- Improves precision in smaller subgroups
- Stabilizes estimates without assuming identical treatment effects

The degree of borrowing is governed by the between-subgroup variance (τ^2), allowing the model to adapt based on the level of heterogeneity observed in the data.

13.4 Control of Type I Error and Multiplicity

Multiplicity arising from multiple endpoints is addressed using a hierarchical testing strategy, with ORR tested prior to PFS.

Although Bayesian methods are used for subgroup analyses, the primary endpoint inference remains frequentist, ensuring alignment with regulatory expectations for Type I error control.

Simulation-based evaluation of the Bayesian model is recommended to confirm that borrowing does not inflate Type I error under heterogeneous treatment effects.

13.5 Risk Assessment and Sensitivity Analyses

Several potential risks are considered in the statistical design:

Heterogeneity Across Subgroups

- Risk: Over-borrowing when treatment effects differ

- Mitigation: Hierarchical model with adaptive τ^2

Missing Data

- Risk: Bias in ORR estimates
- Mitigation:
    - Non-responder imputation (primary)
    - Multiple imputation (sensitivity)
    - Tipping-point analysis

Non-Proportional Hazards (PFS)

- Risk: Misleading hazard ratio interpretation
- Mitigation:
    - RMST as supportive analysis

Treatment Switching

- Risk: Dilution of treatment effect
- Mitigation:
    - Hypothetical estimand

13.6 Clinical and Regulatory Alignment

The overall statistical strategy is designed to:

- Align with ICH E9(R1) estimand principles
- Provide interpretable results for clinicians
- Support decision-making in early-phase development

The integration of frequentist and Bayesian methods allows for both regulatory robustness and improved statistical efficiency, particularly in the context of subgroup analyses.

Appendix 1: Detailed Technical Specifications

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14. Analysis Data Handling and Derivations

14.1 Analysis Data Model (ADaM) Datasets

- **ADSL**: Subject-level dataset containing demographics, treatment assignment, and baseline characteristics
- **ADRS**: Response dataset derived from RECIST tumor assessments
- **ADTTE**: Time-to-event dataset for PFS endpoint

14.2 Key Variable Derivations

- **Objective Response Rate (ORR):**
  ORR = 1 if subject achieves Complete Response (CR) or Partial Response (PR) per RECIST 1.1; otherwise ORR = 0
- **Progression-Free Survival (PFS):**
  PFS time = Date of progression or death − Date of randomization + 1
- **Censoring Rule:**
  Subjects without a PFS event will be censored at the date of last adequate tumor assessment

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14.3 Algorithm Specifications

14.3.1 Response Derivation Algorithm

1. Identify all tumor assessment records
2. Determine best overall response per RECIST 1.1
3. Assign binary response variable (Responder vs Non-responder)

14.3.2 PFS Derivation Algorithm

1. Identify progression date
2. Identify death date
3. Select earliest of progression or death
4. If no event observed:
    - Apply censoring at last adequate assessment